WORKED SOLUTION

STEP I 2003, Question 13

Probability and Statistics • Geometric Distribution Expectation

STEP I 2003, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q13

The key idea

For the first part, count each team's *misses* instead of its goals: after five shots each, the teams are level exactly when they have missed the same number of times. For the expectation, notice that the sudden-death stage happens in rounds of two shots, and every round is an independent trial that ends the shoot-out with the same probability. So the number of rounds is geometric, and the expected number of shots splits into "the ten we always take" plus "the extra ones, weighted by the chance we need them".

Solution

We assume, as the model intends, that every shot is independent of every other shot, and that Team A always scores with probability p_A and Team B with probability p_B. Write $q_A = 1 - p_A$ and $q_B = 1 - p_B$.

The probability α of a draw after ten shots

Let X be the number of shots Team A misses in its first five, and Y the number Team B misses. Each team's five shots are independent trials with the same miss probability, so

$$X \sim \mathrm{B}(5, q_A), \qquad Y \sim \mathrm{B}(5, q_B),$$

and X and Y are independent because the two teams' shots are.

Team A scores $5 - X$ goals and Team B scores $5 - Y$, so the scores are level exactly

when $X = Y$. The events $\{X = Y = i\}$ for $i = 0, 1, \ldots, 5$ are disjoint, so

$$\begin{aligned}\alpha = \mathrm{P}(X = Y) &= \sum_{i=0}^{5} \mathrm{P}(X = i)\,\mathrm{P}(Y = i) \\ &= \sum_{i=0}^{5} \binom{5}{i} q_A^i p_A^{5-i} \binom{5}{i} q_B^i p_B^{5-i} \\ &= \sum_{i=0}^{5} \binom{5}{i}^2 (1-p_A)^i (1-p_B)^i \, p_A^{5-i} p_B^{5-i},\end{aligned}$$

which is the required expression. (Counting goals instead, with $5 - i$ goals each, gives the same sum, since $\binom{5}{i} = \binom{5}{5-i}$.)

The expected number of shots

If the scores are still level after ten shots, the sudden-death stage runs in *rounds*: in each round A takes one shot and B takes one shot. A round settles the match when exactly one of the two shots goes in, and that happens with probability

$$p_A q_B + q_A p_B = p_A(1 - p_B) + p_B(1 - p_A) = p_A + p_B - 2p_A p_B = \beta.$$

Otherwise (both score or both miss) another round is played. We need $\beta > 0$, that is, we exclude the trivial cases $p_A = p_B = 0$ and $p_A = p_B = 1$ in which no round can ever be decisive.

Ask yourself Why can't the shoot-out stop halfway through a round?

The rules say the teams must have taken the same number of shots when one has scored and the other hasn't. Even if A misses, B still has to take its shot (B might miss too). So the sudden-death shots always come in pairs, and the total is $10 + 2R$, where R is the number of rounds played ($R = 0$ if the match is settled in the first ten shots).

Given that sudden death is reached, the rounds are independent trials, each decisive with probability β, so R has a geometric distribution:

$$\mathrm{P}(R = k \mid \text{level after ten}) = (1 - \beta)^{k-1}\beta, \qquad k = 1, 2, 3, \ldots$$

Its mean is

$$\mathrm{E}(R \mid \text{level after ten}) = \sum_{k=1}^{\infty} k\beta(1 - \beta)^{k-1} = \beta \cdot \frac{1}{(1 - (1 - \beta))^2} = \frac{1}{\beta},$$

using $\sum_{k \geq 1} k x^{k-1} = (1 - x)^{-2}$ for $|x| < 1$, which applies because $0 < \beta \leq 1$.

If the match is settled in the first ten shots, $R = 0$. Conditioning on whether the scores are level after ten shots,

$$\mathrm{E}(R) = \alpha \cdot \frac{1}{\beta} + (1 - \alpha) \cdot 0 = \frac{\alpha}{\beta}.$$

Hence the expected number of shots is

$$\mathrm{E}(10 + 2R) = 10 + \frac{2\alpha}{\beta},$$

as required.

Checks

- Equal teams with $p_A = p_B = \frac{1}{2}$: by Vandermonde's identity $\sum_i \binom{5}{i}^2 = \binom{10}{5} = 252$, so $\alpha = 252/2^{10} = \frac{63}{256}$, and $\beta = \frac{1}{2}$. The expected number of shots is $10 + \frac{63}{64} = \frac{703}{64} \approx 10.98$.
- If one team is perfect and the other hopeless ($p_A = 1$, $p_B = 0$), then $\alpha = 0$ (only the $i = 0$ term could survive, and it contains $p_B^5 = 0$), so exactly 10 shots are taken, as common sense says.
- We checked the formula for α against a direct enumeration of all 2^{10} outcomes of the first ten shots, and $10 + 2\alpha/\beta$ against a simulation of 2×10^5 shoot-outs; for $p_A = \frac{3}{4}$, $p_B = \frac{2}{3}$ the formula gives 11.264 and the simulation 11.25.

Common slips

- Taking β to be the probability that A wins a round, $p_A(1 - p_B)$: the shoot-out ends when *either* team wins a round.
- Writing the answer as $10 + 2/\beta$, forgetting that sudden death happens only with probability α.
- Counting one shot per step of sudden death instead of two.

Beyond STEP

The step $\mathrm{E}(R) = \alpha\,\mathrm{E}(R \mid \text{level}) + (1 - \alpha)\,\mathrm{E}(R \mid \text{not level})$ is the law of total expectation, one of the most useful tools in first-year probability. The sudden-death stage is also a simple Markov chain with an absorbing state, and the "expected time to absorption" calculation here generalises to random walks and gambler's-ruin problems. Grimmett and Welsh, *Probability: An Introduction*, covers both conditional expectation and these chains.

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